Week #11: 9 Nov - 13 Nov

Subject/:الموضوع Chemistry

Unit Title: عنوان الوحدة **Unit 4: Oxidation and reduction**

Grade الصف: 10

Lessons الحصص	Learning targets أهداف التعلم	Assessment tools and strategies التقييم Formative تكويني/ Summative ختامي / Differentiation التمايز	Resources and links for Parents و المصادر الروابط المخصصة لأولياء الأمور
Lesson 1: Batteries	**I can EXPLAIN how batteries use redox reactions to convert chemical energy into electrical energy and COMPARE primary and secondary batteries.**	1.1 Flexible display of information (size, color, font, layout). 2.1 Clarify vocabulary and symbols using visuals, connections, and supports. 4.2 Ensure access to tools and assistive technologies.	Lesson 1: McGraw-Hill Unit 4, Batteries PPT, Chemistry Notebook. Lesson 2: Formative revision resources, GRASP task brief and rubric. Resources on Teams and Toddle.
Lesson 2: Formative & GRASP Progress Check	I can APPLY my understanding of batteries and redox reactions in a formative assessment and EVALUATE my GRASP project progress using the task criteria and feedback.		
Homework الواجبات	Review batteries, oxidation and reduction using Teams resources. Continue the GRASP project using the task checklist.		
Teacher's notes/ Reminders ملاحظات المعلم والتذكير	Hello Parents, Please encourage your child to review and practice Science regularly each week using the resources available in their class Microsoft Teams - class material folder which is organized by week. The presentations, readings, videos, practice activities, etc. posted in Teams are the primary learning materials and should be used throughout the unit to support understanding and achievement. Science study guides will provide basic *assessment information, topics, and expectations*; however, they are intended as a guide only and will not contain detailed study materials, these are found in the class teams. Effective preparation for graded formatives and summative requires students to actively engage with the learning materials provided in their Toddle class drive. Thank you for partnering with us to support your child's learning and success. Ms. Stacey (Science Coordinator)		

Technology Integration دمج التكنولوجيا	Battery diagrams and digital models illustrate redox and energy conversion. Teams/Toddle support revision, GRASP resources and feedback.
Cross-curricular links الروابط بين المواد الدراسية	**Physics: electrical energy, charge flow and circuits.** **Mathematics: interpreting data and comparing battery performance.** **Environmental Science: responsible battery use and recycling in the UAE.**
External Assessment Integration دمج التقييمات الخارجية	MAP , CAT 4
Wellbeing/ ATL Skills الصحة النفسية/ أساليب التعلم	Critical Thinking: explain battery redox and justify choices. Information Literacy: select reliable evidence for GRASP. Organization: track project milestones and next steps. Collaboration: provide constructive peer feedback. Communication: explain scientific ideas and progress clearly.
Standards المعايير	**HS-PS1-2: Construct and revise explanations of chemical reactions using electron states and periodic trends.**
Inquiry/ TOK Question البحث والاستقصاء أسئلة TOK	**How do redox reactions enable batteries to produce electricity, and how can scientific feedback improve our GRASP work?**
Success Criteria معايير النجاح **(Measurable)**	L1 – BATTERIES MUST (DOK 1–2): IDENTIFY the anode, cathode and electron flow in a battery. **SHOULD (DOK 3): EXPLAIN how redox reactions generate electricity and COMPARE primary and secondary batteries.** COULD (extension): EVALUATE a battery choice for a real-life UAE application using scientific evidence. L2 — FORMATIVE & GRASP PROGRESS CHECK **MUST (DOK 1–2): RECALL key redox and battery concepts and IDENTIFY completed GRASP requirements.** SHOULD (DOK 3): APPLY scientific understanding in the formative and ANALYZE GRASP progress using the rubric.

	COULD (extension): JUSTIFY project improvements and DEVELOP a prioritized next-step action plan.
National Identity Domain مجالات الربط بالهوية الوطنية	Conservation | Global Understanding | UAE Values & Sustainable Development **Connect battery technologies to sustainable energy use and responsible disposal/recycling in the UAE.** **During the GRASP check, students consider how their scientific solutions affect the UAE community and environment.**